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**EX NO: 10 CALCULATION OF PROCESSING GAIN AND GENERATION OF
PN SEQUENCES FOR SPREAD SPECTRUM COMMUNICATION**

PROGRAM (MATLAB Code)

```
clc;
clear;
close all;
% PN sequence lengths
PN_length = [7 15 31 63 127 255 511 1023];
% Processing Gain (Linear)
PG_linear = PN_length;
% Processing Gain in dB
PG_dB = 10*log10(PG_linear);
% Display table
T = table(PN_length',PG_dB',...
'VariableNames',{'PN_Length','Processing_Gain_dB'});
disp('Processing Gain Table');
disp(T);
% Plot
figure;
stem(PN_length,PG_dB,'filled');
grid on;
xlabel('PN Sequence Length');
ylabel('Processing Gain (dB)');
title('Spread Spectrum: Processing Gain vs PN Sequence Length');
```

Objective 1:

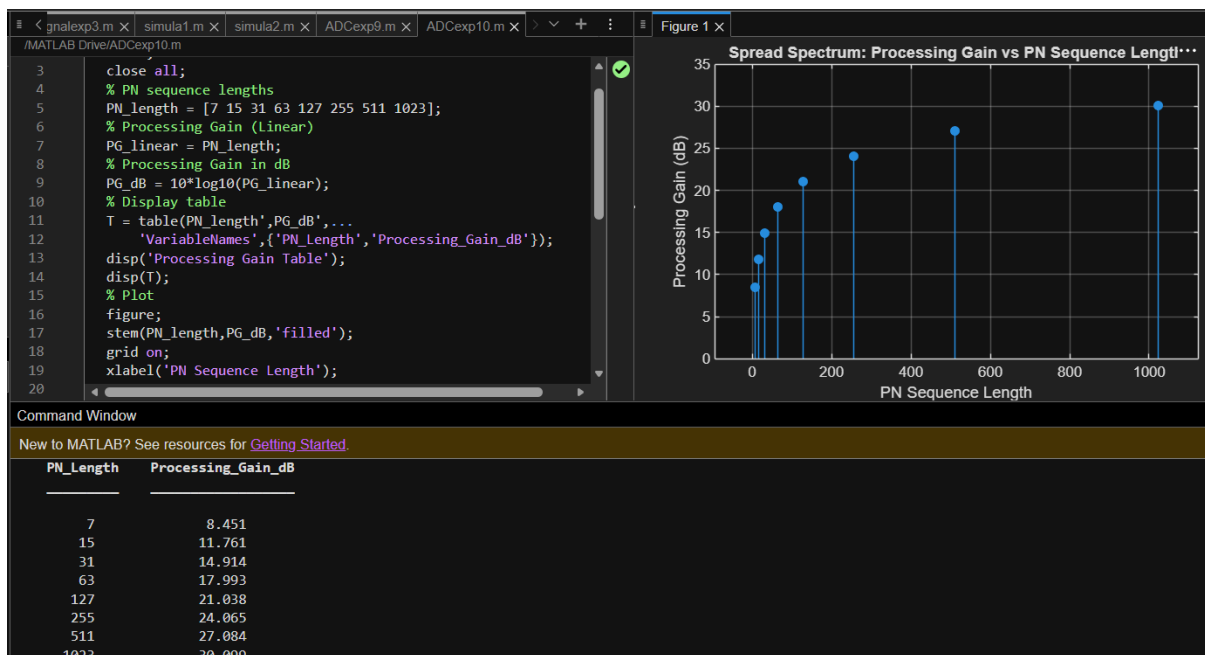
To calculate the processing gain for various PN sequence lengths and tabulate the corresponding

values.

Objective 2:

To visualize and compare the processing gain obtained for different PN sequence lengths using MATLAB plots.

OUTPUT



Result

The MATLAB plot successfully illustrated the relationship between PN sequence length and processing gain. Longer PN sequences produced higher processing gains, thereby improving interference rejection and communication reliability..